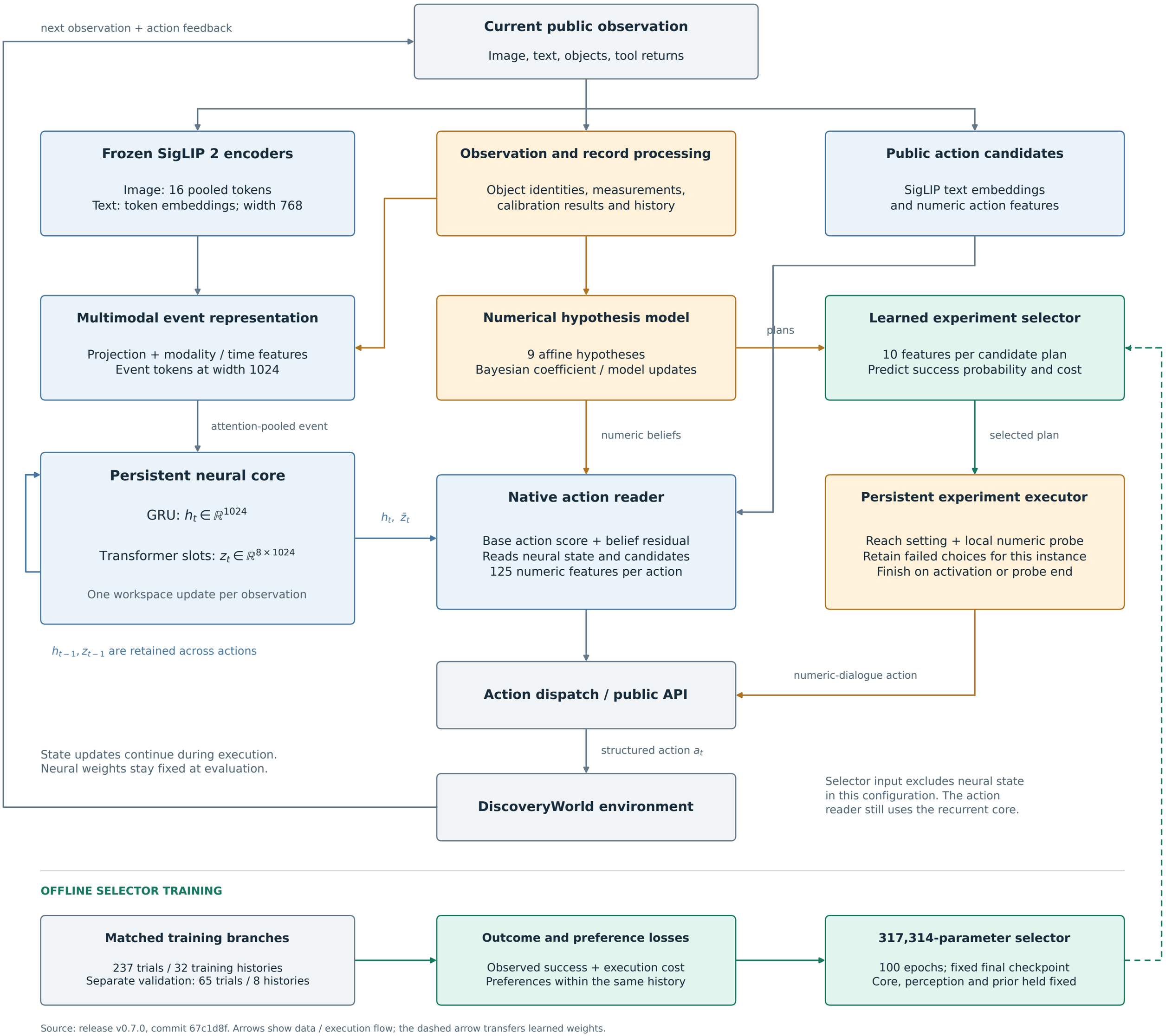
Continual Decision Model

v0.7.0 | Deployed numerical-selector configuration | Reactor Lab

Weights fixed in this study

Trained in v0.7.0

Explicit state / numerical procedure

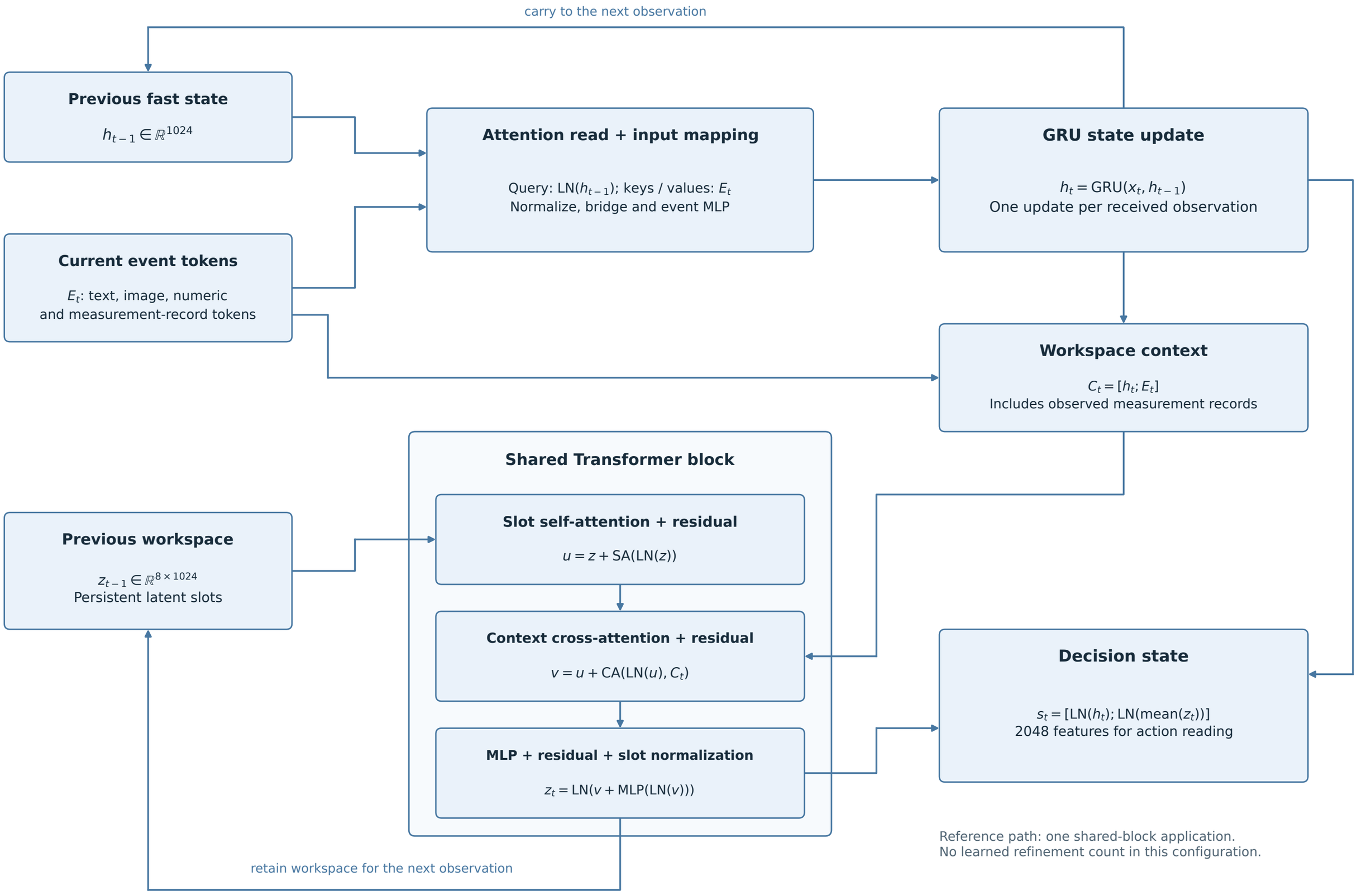


OFFLINE SELECTOR TRAINING

Source: release v0.7.0, commit 67c1d8f. Arrows show data / execution flow; the dashed arrow transfers learned weights.

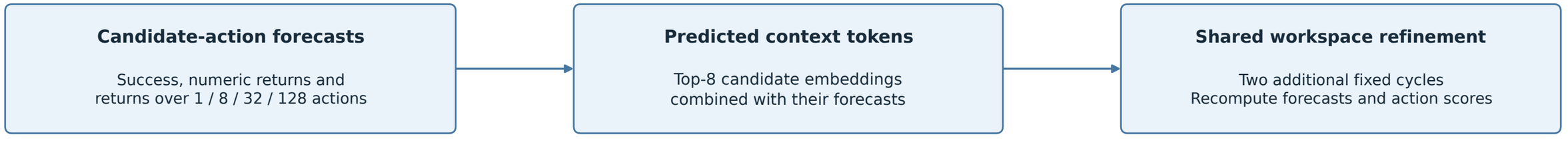
Persistent core: one observation update

Exact computational structure of the reference-reader path | Width 1024; eight latent slots



Reference path: one shared-block application.
No learned refinement count in this configuration.

SEPARATE PREDICTIVE-PARENT CONFIGURATION



This prediction-feedback path is not used by the transferred numerical selector that completed 23/24 instances.

Where computation, memory and learning reside

Architecture comparison | Schematics redrawn from code and primary papers

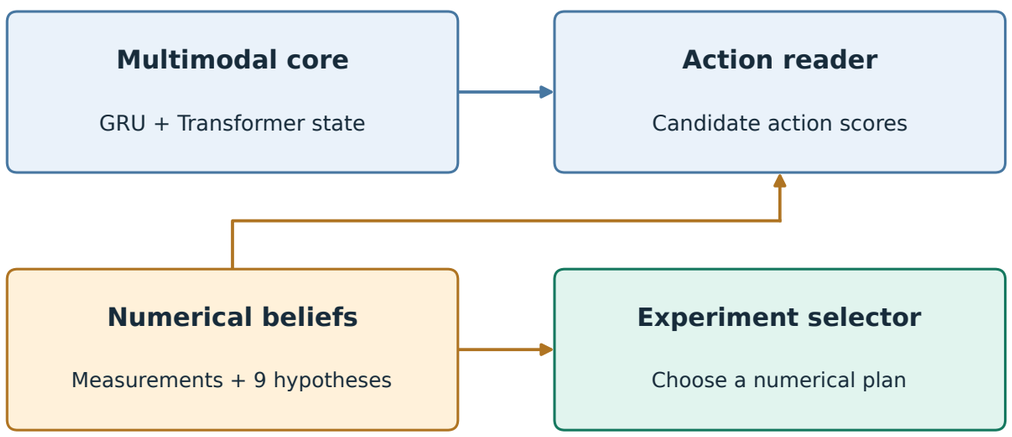
Neural module / representation

Learned or adapted component

Explicit state / numerical procedure

a CDM v0.7.0

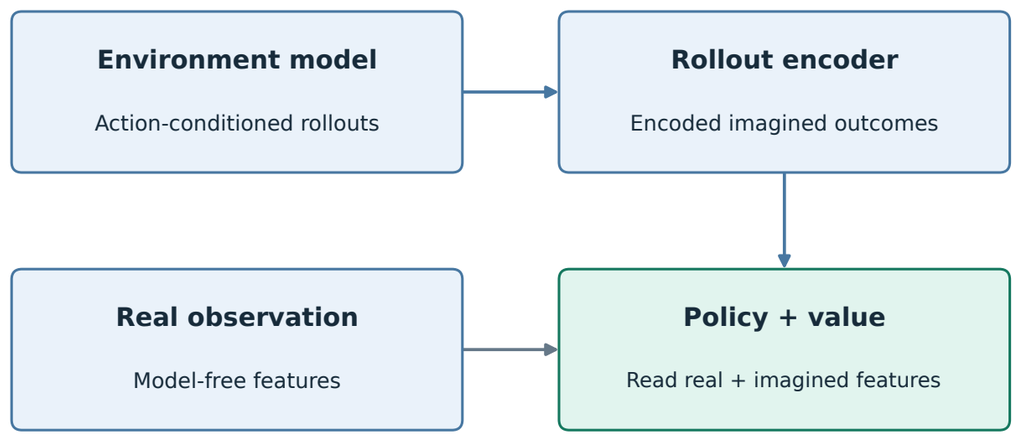
Numerical-selector configuration; native action execution



Runtime: h / z, numerical posteriors and plan state update.

b I2A [1]

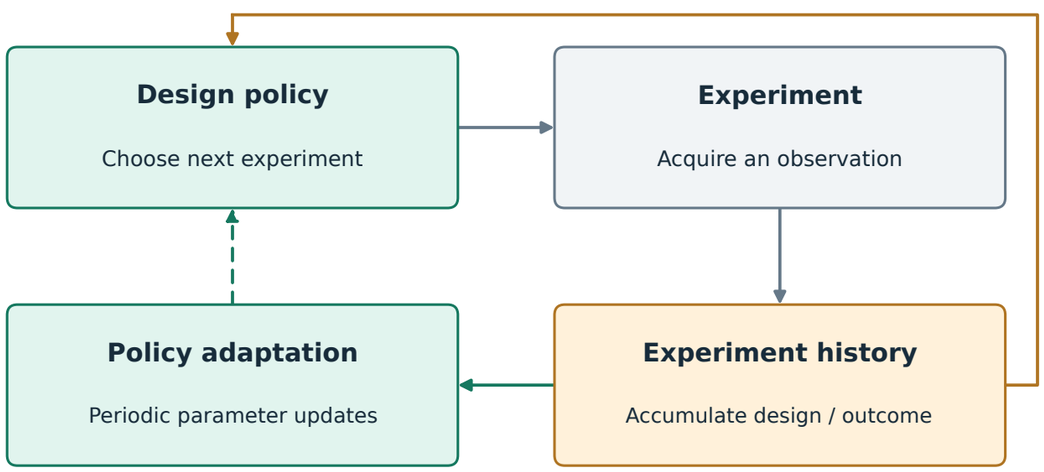
Learned interpretation of model-generated futures



Predicted trajectories provide context for action selection.

c Step-DAD [2]

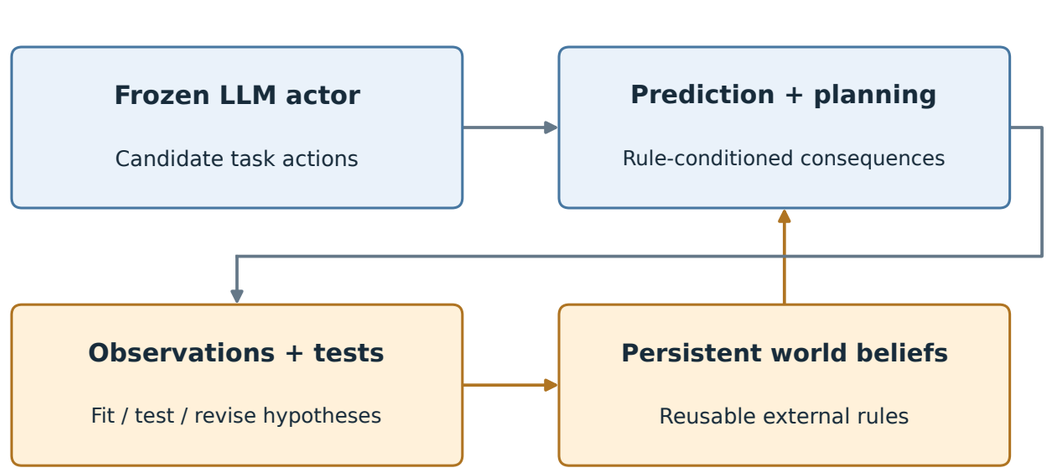
Sequential experimental design with test-time adaptation



Observed data refine the deployed experimental-design policy.

d EvoBelief [3]

Frozen language model with an evolving external world model



Rules change during deployment; LLM weights remain fixed.

READING THE COMPARISON

CDM maintains a neural state and explicit numerical state; v0.7.0 learns the experiment selector offline.

Its prediction-feedback variant is a separate configuration. These diagrams compare mechanisms, not benchmark rankings.

[1] Weber et al., NeurIPS 2017, arXiv:1707.06203. [2] Hedman et al., ICML 2025, PMLR 267.

[3] Wang et al., 2026, arXiv:2610.12789 (preprint). Schematic connections omit task-specific implementation details.